RAK13101 WisBlock GSM/GPRS Module Datasheet

Overview

Description

WisBlock RAK13101 GSM/GPRS Module is a part of the WisBlock Wireless series that provides GSM/GPRS capability on the WisBlock platform by using Quectel MC20CE cellular module.

Features

- Quectel MC20CE module
- Supports GSM/GPRS/GNSS: 850/900/1800/1900 MHz
- Supports BeiDou/GPS/GLONASS/QZSS
- Built-in LNA
- IPEX connectors for the GSM and GNSS antenna
- Micro-USB debug and log output connector
- Nano SIM support
- 3.3 V Power supply
- Small size: 25 mm x 45 mm

Specifications

Overview

The RAK13101 module can be mounted on the IO slot of the WisBlock Base board. Figure 1 shows the mounting mechanism of the RAK13101 on a WisBlock Base module, such as a RAK5005-O.

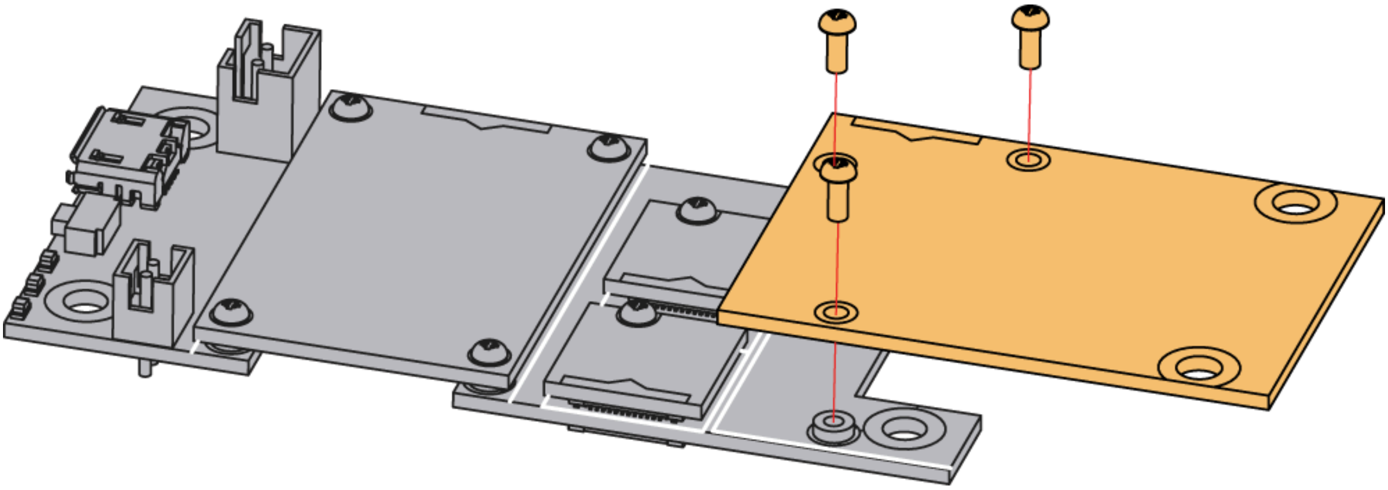

Figure 1: RAK13101 mounting mechanism on a WisBlock Base module

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts and its corresponding functions and diagrams. It also covers the electrical and mechanical parameters that include the tabular data of the functionalities and standard ratings of the RAK13101 WisBlock GSM/GPRS Module.

Chipset

Vendor	Part number
Quectel	MC20CE

Pin Definition

The RAK13101 WisBlock GSM/GPRS module comprises a standard WisBlock IO slot connector. The WisBlock connector allows the RAK13101 module to be mounted on a WisBlock baseboard with IO slot, such as RAK5005-O. The pin order of the connector and the pinout definition is shown in **Figure 2**.

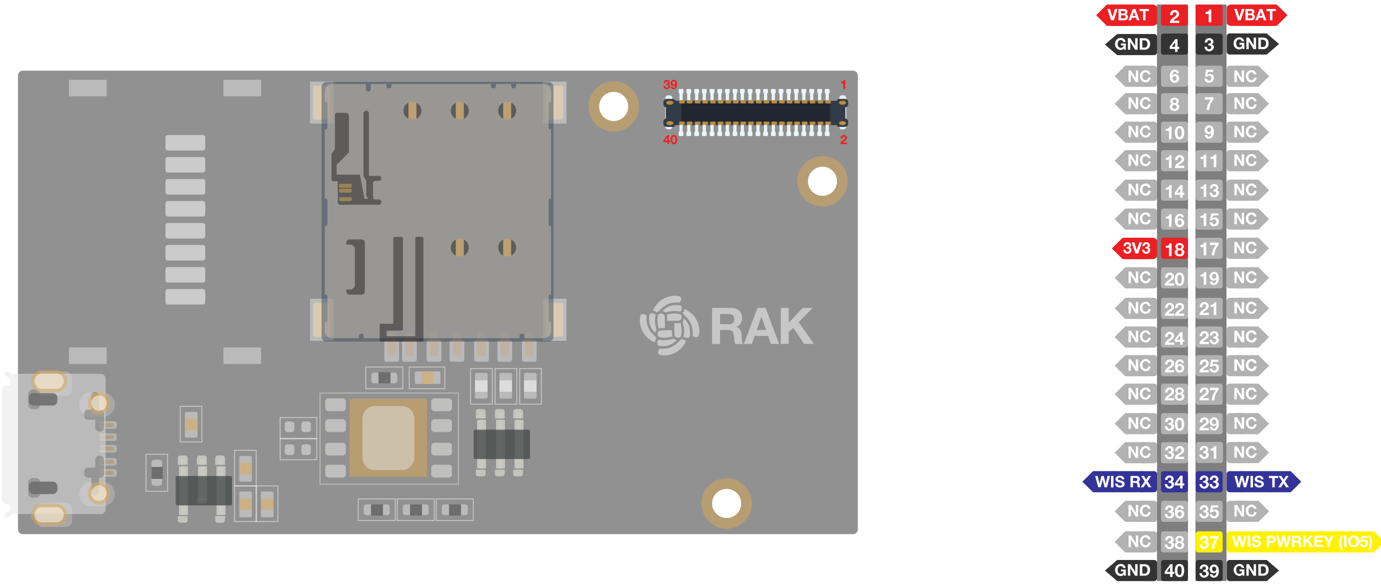

Figure 2: RAK13101 Connector Pin Definition

NOTE

- RAK13101 WisBlock IO slot connector utilizes the **UART** related pins, **PWRKEY** via WB_IO5, **VBAT**, **3V3**, and **GND**.
- **VBAT** is the battery voltage input with max voltage 4.2 V. During GPRS data and GSM dial mode, the peak current is 1170 mA and 769 mA, respectively, which exceeds the USB port supply current. That is why you must use a dedicated battery.

Electrical Characteristics

Absolute Maximum Ratings

Parameter	Minimum	Maximum	Unit
VBAT	-0.3	+4.2	V
GNSS_VCC	-0.3	+4.2	V
Power supply peak current0	0	2	A

Power Supply Ratings

Symbol	Description	Condition	Min.	Nom.	Max.	Unit
VBAT	Supply Voltage	Input voltage must within this range	3.3	4.0	4.2	V
GNSS_VCC	GNSS Part Supply voltage	Input voltage must within this range	2.8	3.3	4.2	V
Ipeak	Peak Current @ VBAT = 3.8 V	Idle	-	-	110	mA
	Peak Current @ VBAT = 3.8 V	GSM Dial	-	-	769	mA
	Peak Current @ VBAT = 3.8 V	GPRS Data	-	-	1170	mA
Ivbat	Average Supply Current	Power down mode	-	20	-	mA
	Average Supply Current	Sleep mode @DRX = 5	-	1.2	-	mA

Mechanical Characteristics

Board Dimensions

Figure 3 shows the dimensions and the mechanical drawing of the RAK13101 module.

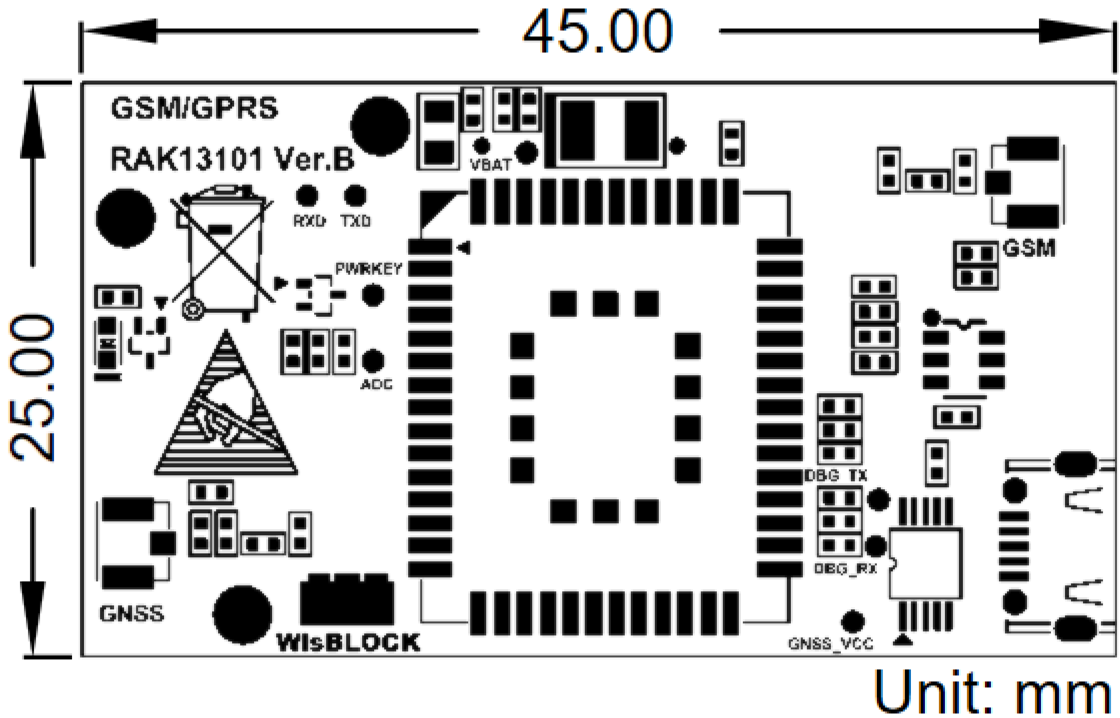

Figure 3: RAK13101 Mechanical Drawing

WisConnector PCB Layout

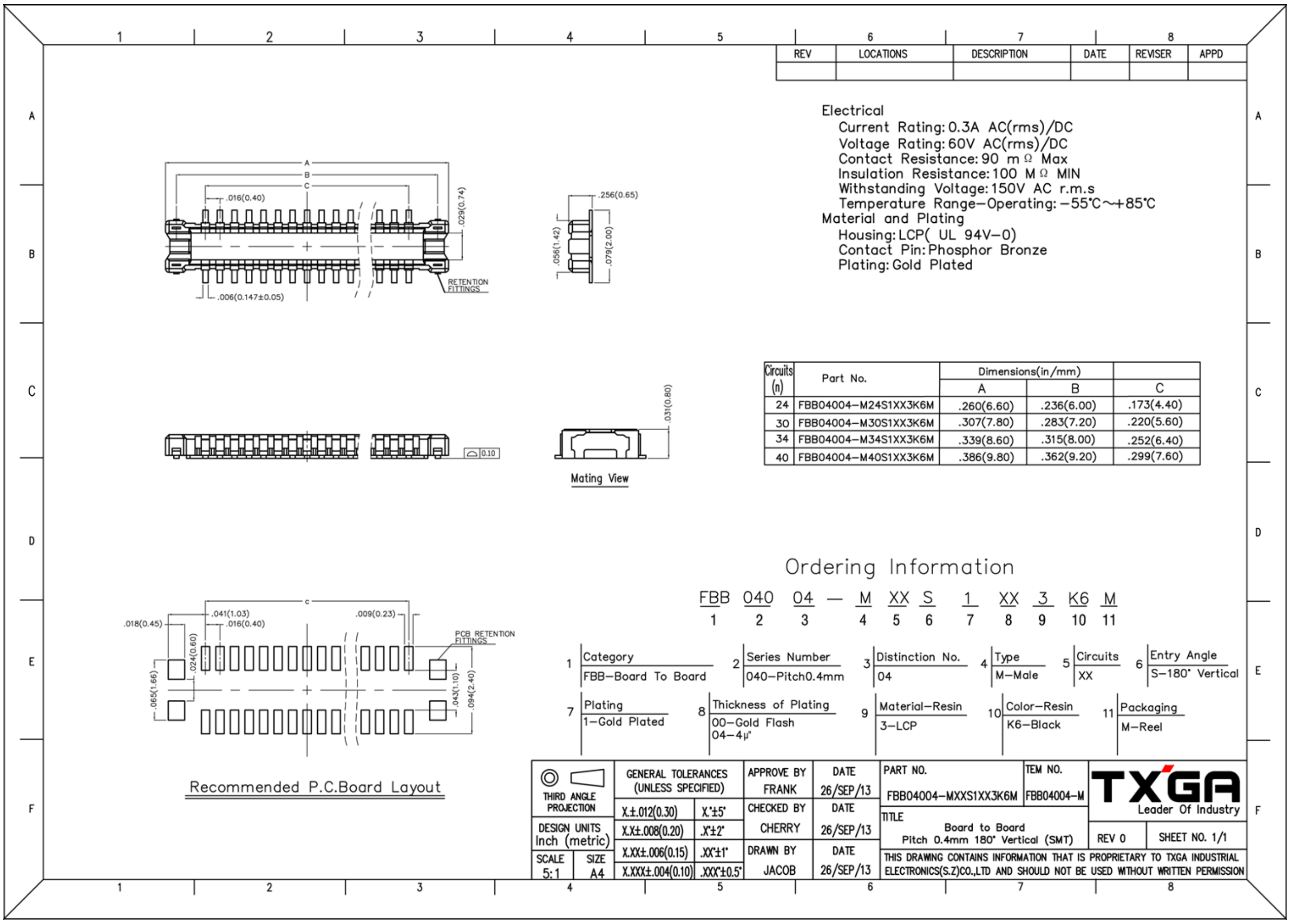

Figure 4: WisConnector PCB Footprint and Recommendations

Schematic Diagram

Power Supply

The Quectel MC20CE module's main power supply comes from **VBAT**, which is a battery voltage connected to the WisBlock Base board. **GNSS_VCC** is the GNSS section supply voltage and is turned on or off via **GNSS_VCC_EN**, which is connected to MC20CE. The power supply of the GNSS part is controlled via the AT command `AT+QGNSSC`.

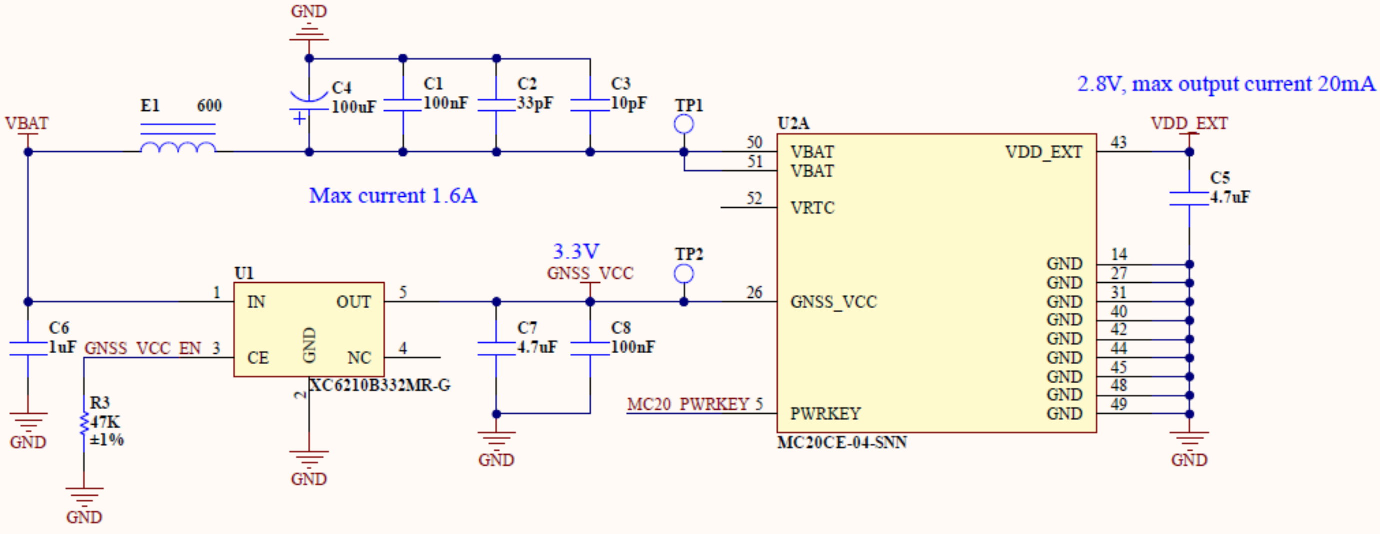

Figure 5: RAK13101 Power Supply

GSM and GNSS

J3 is GSM antenna connector and **J2** is GNSS antenna connector.

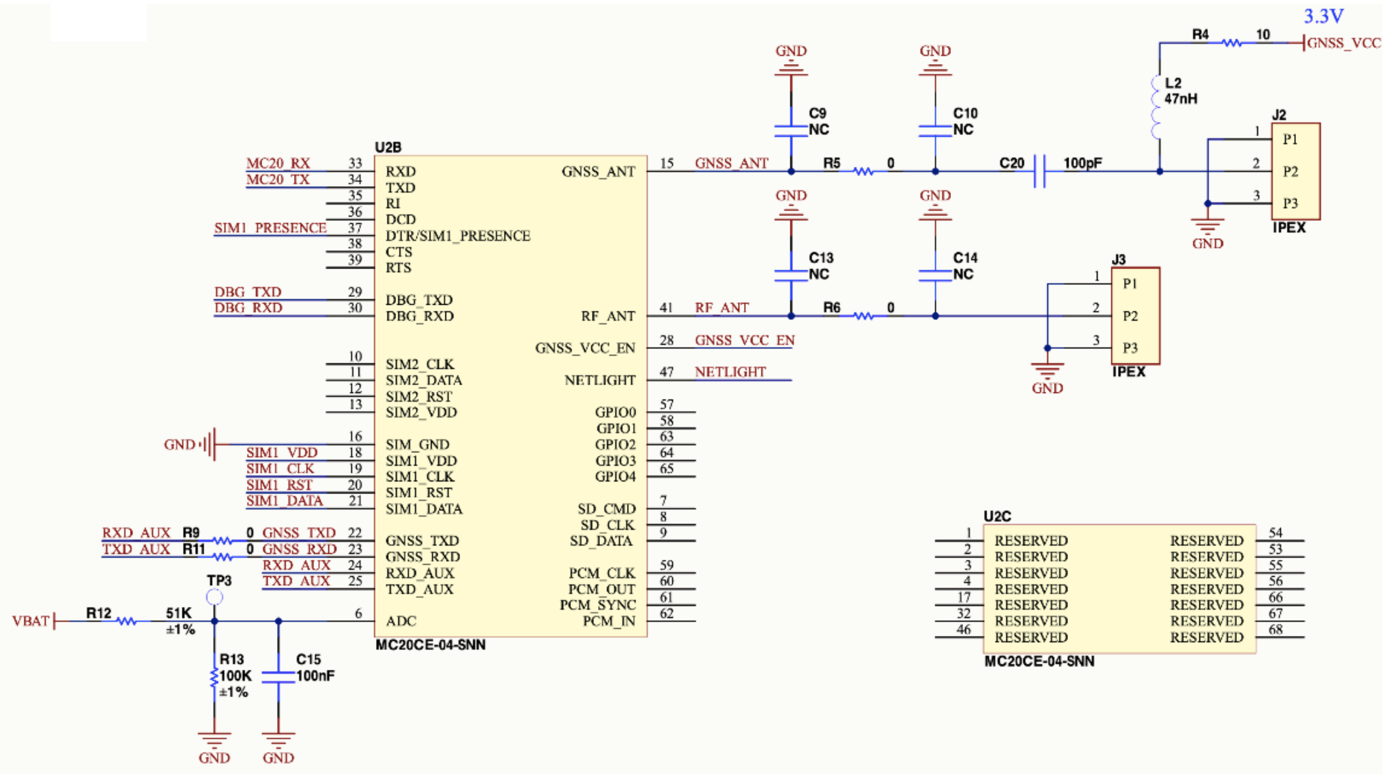

Figure 6: RAK13101 GSM and GNSS Antenna Circuit

Voltage Level Transfer

The GSM/GPRS module operates at 2.8 V, while the WisBlock Core is at 3.3 V. That is why a voltage level shifter is needed.

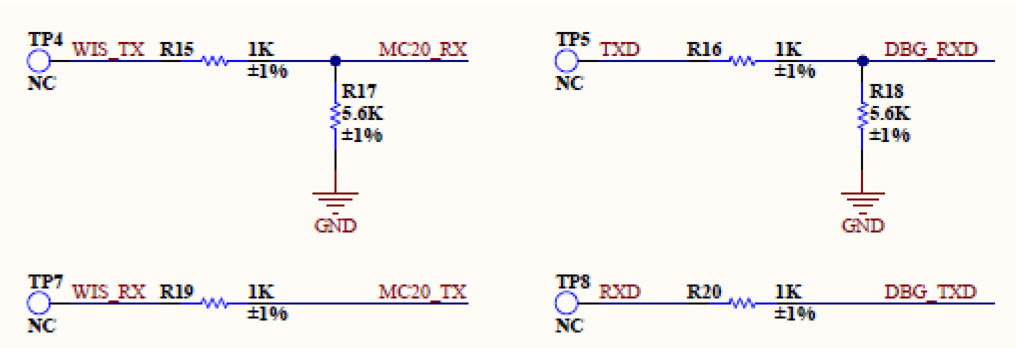

Figure 7: RAK13101 Voltage Level Shifter Circuit

SIM Circuit

Figure 8 shows the schematic of RAK13101 module as a slot for standard SIM card.

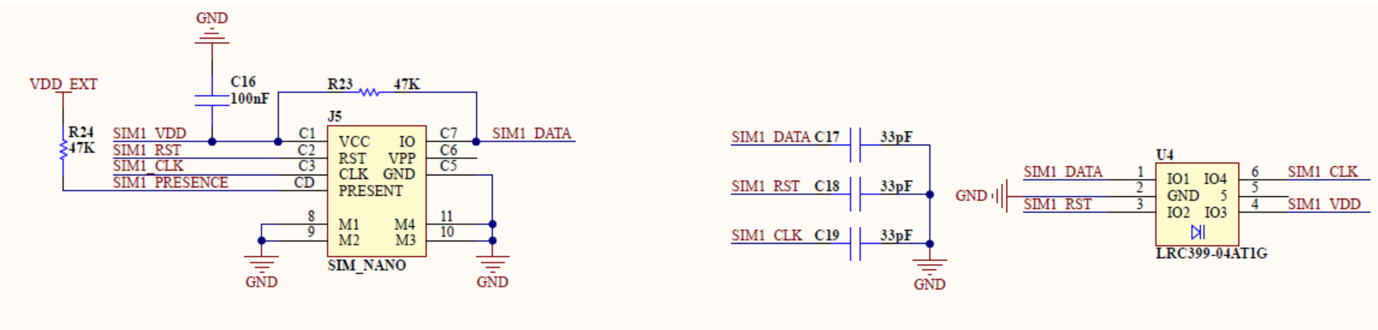

Figure 8: RAK13101 GSM/GPRS Module SIM Circuit

WisConnector

RAK13101 WisBlock IO slot connector utilizes **UART** related pins, **PWRKEY** via WB_IO5, **VBAT**, **3V3**, and **GND**.

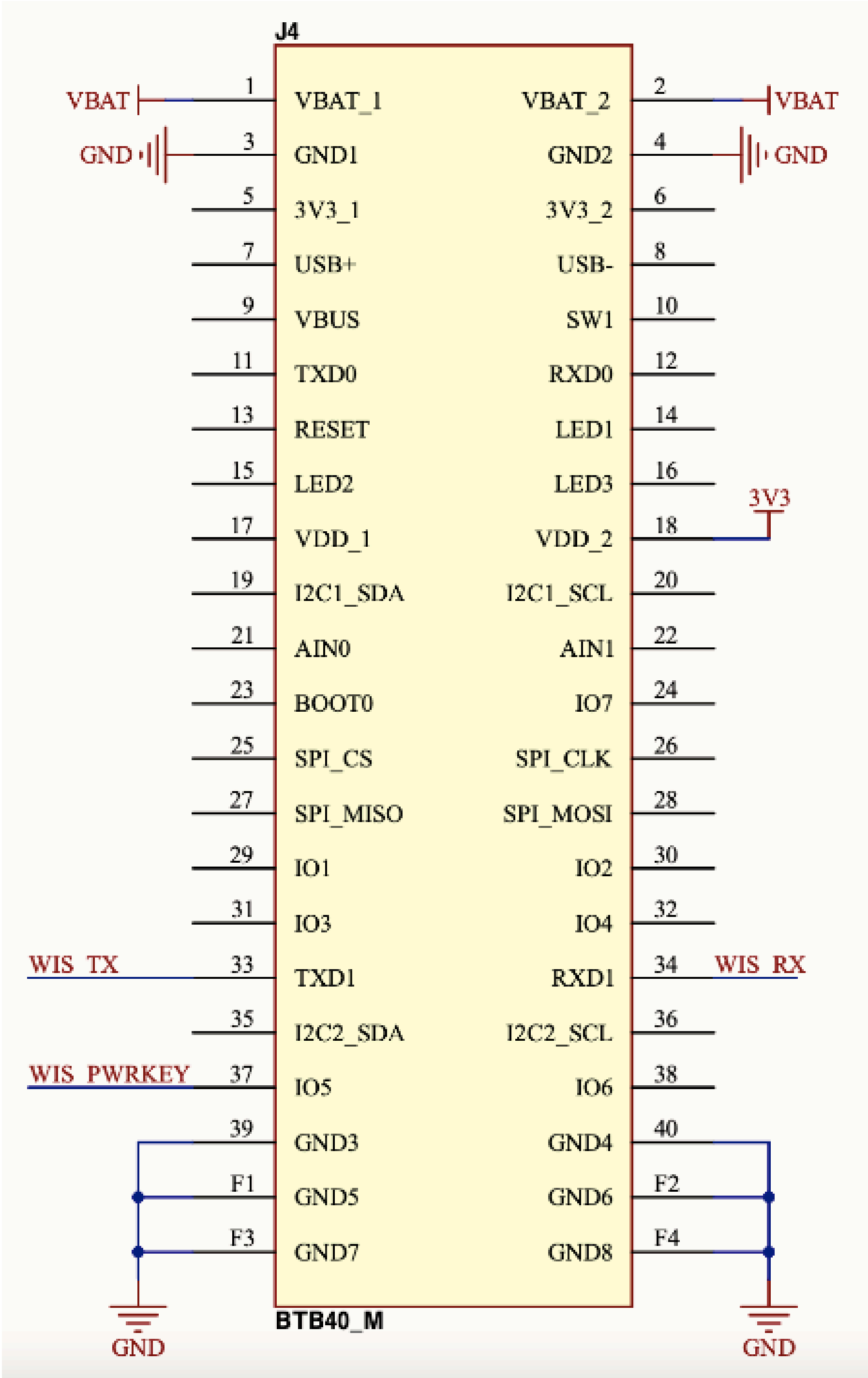

Figure 9: RAK13101 IO Slot Connector

Debug Connector

Figure 10 shows the RAK13101 USB debugging circuit.

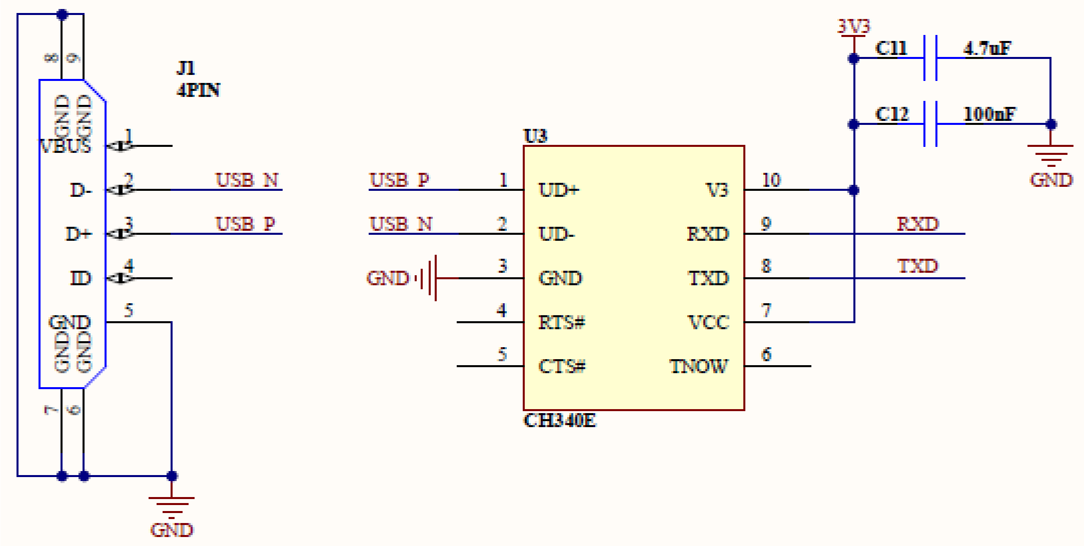

Figure 10: RAK13101 USB Debugging

LED/PWRKER Control Circuit

Figure 11 shows the RAK13101 module LED and PWRKEY controlled circuit.

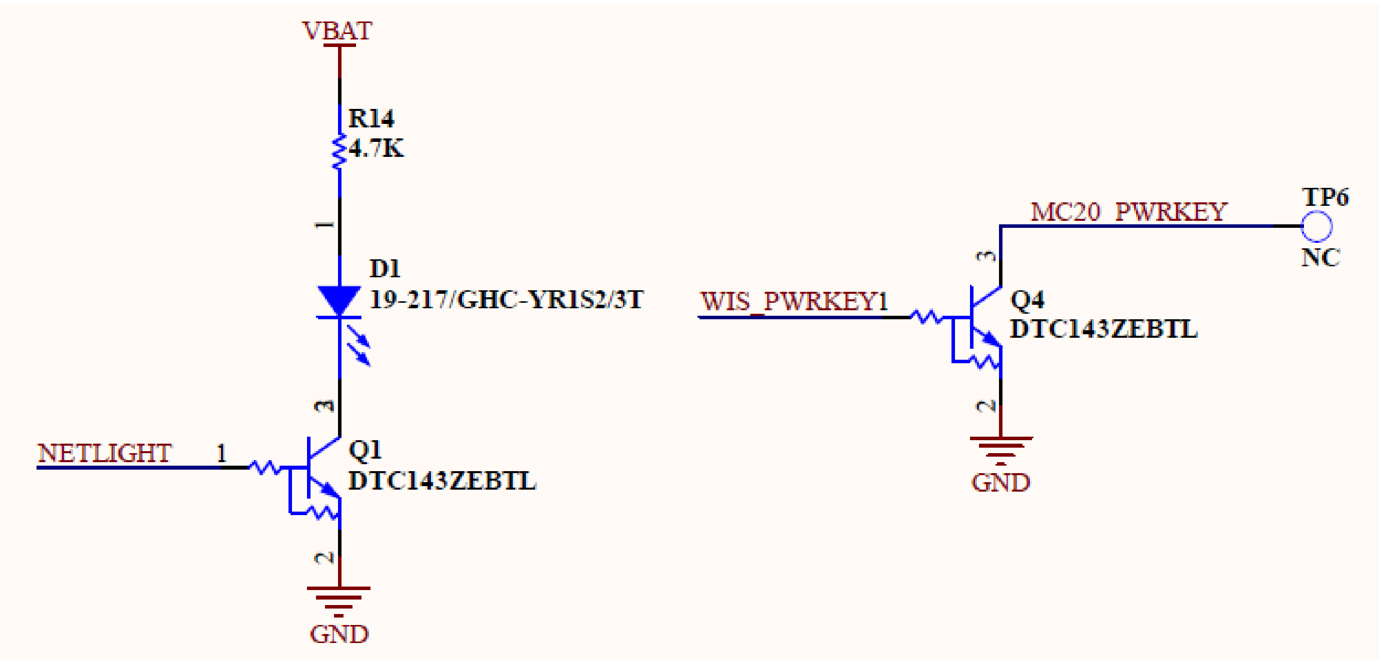

Figure 11: RAK13101 Module LED PWRKEY Control Circuit